



AgriGaurd smart irrigation & smart water management

Smarter water. Better irrigation. Sustainable farming.



Problem

Small-scale farmers face challenges such as:

1. Limited water availability
2. Inefficient irrigation practices
3. Over-irrigation and water wastage
4. Different soil types having different water requirements
5. Irrigation decisions often relying on fixed schedules
6. Limited access to affordable smart-farming technology

Question: How can we help farmers use limited water more efficiently while still providing crops with the water they need?

Solution: AgriGaurd

AgriGuard uses **real-time sensing + automated control + water management** to make irrigation decisions.

Zone-Based :
Each area can be monitored and irrigated independently.

Adaptive :
The system records drying rates and irrigation cycles.

Water-aware :
Available tank water is considered when allocating irrigation.

Weather-Assisted :
Weather information can improve irrigation decisions.

Safe :
Physical E-STOP and reset controls are incorporated.

Why can AgriGuard become a real product?



Marketability

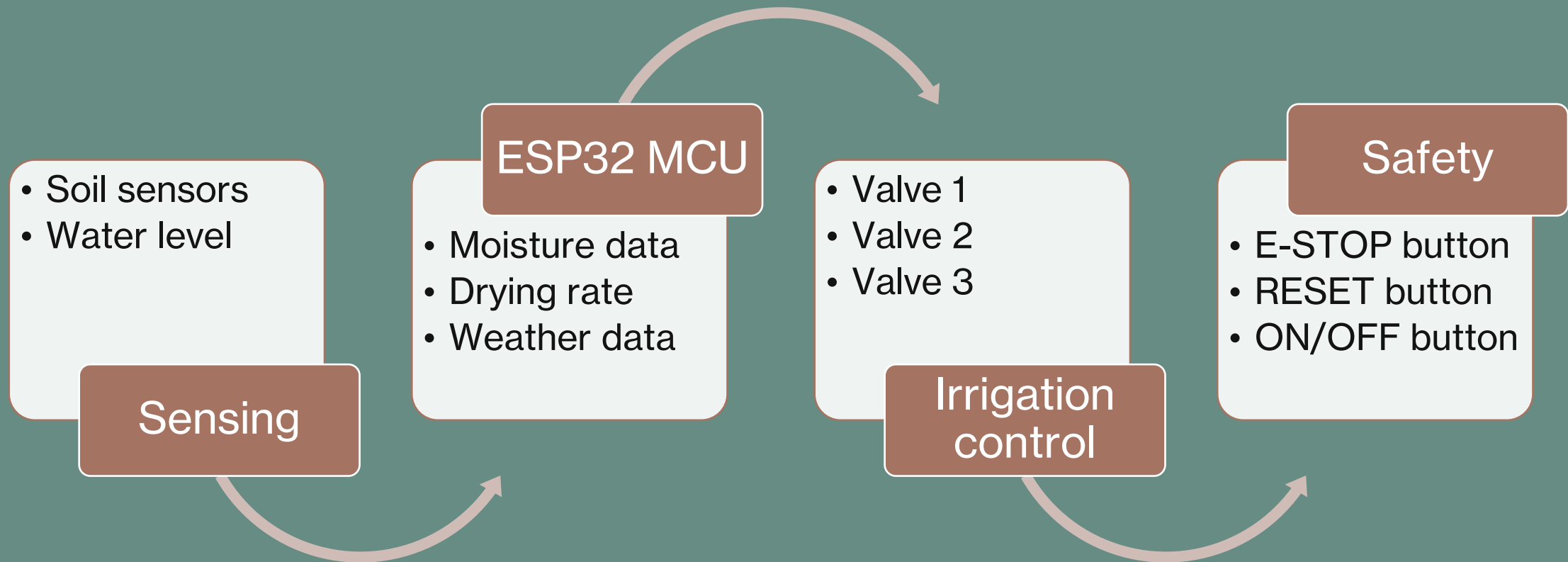
- ✓ Vegetable growers
- ✓ Small-scale farmers
- ✓ Community gardens
- ✓ Small agricultural operations
- ✓ Water-conscious growers



Buildability

- ✓ The prototype uses commercially available components:
- ✓ ESP32
- ✓ Capacitive soil sensors
- ✓ Relay module
- ✓ Solenoid valves
- ✓ Water-level sensor
- ✓ LCD
- ✓ Standard switches and resistors

System & Technical Design



Bill of components

Components	Quantity	Estimated cost
<u>ESP32 Dev board</u>	1	R 220.00
<u>Capacitive soil sensor</u>	3	R 660.00
<u>Ultrasonic sensor</u>	1	R 22.00
<u>Solenoid irrigation valve</u>	3	R 276.00
<u>4-channel relay module</u>	1	R 50.00
<u>LCD I2C board</u>	1	R 103.00
<u>ON/OFF latch switch</u>	2	R 80.00
<u>E-STOP latching switch</u>	1	R 86.00
<u>Pull up resistors PACK (2-4KΩ)</u>	1	R 100.00
<u>3D printed Encapsulator(Fillament)</u>	1	R 200.00
<u>Tubing/piping (5m)</u>	1	R 130.00
Wiring connectors (5m)	3	R 78.00
<u>Flow sensor</u>	1	R 69.00
12V Battery	1	R 750.00
Total :		R 2824.00

AgriGaurd team lineup



What does this project need:
GOD & Alfred 🤖 👍



Project Manager

- Oversees project architecture and system integration
- Coordinates team activities and project milestones



Project Researcher

- Researches suitable sensors and electronic components
- Evaluates component performance, cost, and suitability



Electronic & Mechanical Designer

- Designs electronic circuits and system wiring
- Develops mechanical components and enclosure designs



Embedded Systems Engineer

- Programs the ESP32 firmware
- Develops sensor-processing and irrigation algorithms



Project Documentation

- Maintains technical documentation and project reports
- Documents system designs, testing, and development
- Prepares technical presentations and supporting materials